FlipVQA: Scaling Multi-modal Instruction Tuning via Textbook-to-Knowledge Synthesis

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Unpublished working draft. Not for distribution.

Abstract

Textbooks are among the richest repositories of human-verified reasoning knowledge, yet their complex layouts contain multi-column typesetting, cross-page question answer separation, and interleaved figures, make automated extraction of structured QA and VQA pairs extremely challenging. Existing alternatives either synthesize data from scratch, which lacks authentic problem contexts, or rely on costly expert annotation that cannot scale. We propose **FlipVQA-Miner**, an automated pipeline that resolves long-range logical dependencies and cross-page discontinuities in OCR-parsed documents, recovering coherent question–answer–figure associations even when answers reside in separate companion volumes. A subsequent multi-stage curation pipeline transforms these raw extractions into training-ready supervision signals. Using FlipVQA-Miner, we construct **FlipVQA-85K**, comprising 85K QA and VQA pairs spanning 11 academic disciplines, at a **50**× cost saving compared to manual annotation while maintaining high structural fidelity ($F_1 > 0.96$). Models fine-tuned on FlipVQA-85K demonstrate significantly improved reasoning ability and cross-domain generalization, establishing a scalable paradigm for human-knowledge-grounded data curation. Our dataset and the complete data generating and curating methods can be found in https://github.com/OpenDCAI/DataFlow-VQA.

CCS Concepts

• **Computing methodologies** → *Image and video acquisition.*

Keywords

Data Mining & Curation, Multimodal Dataset Generation, Multimodal Reasoning

ACM Reference Format:
Zhen Hao Wong, Jingwen Deng, Yuzhao Wang, Wenkai Yu, Jihao Huang, Runming He, Chengyu Shen, Hao Liang, and Wentao Zhang. 2018. FlipVQA: Scaling Multi-modal Instruction Tuning via Textbook-to-Knowledge Synthesis. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX).* ACM, New York, NY, USA, 23 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

The reasoning capability of Large Language Models (LLMs) has become central to a wide range of applications, including complex problem solving, mathematical theorem proving, and multimodal understanding [7, 22]. As modern models increasingly rely on reasoning ability, the availability of high-fidelity, reasoning-oriented multimodal data has emerged as a critical bottleneck.

Textbooks constitute one of the most valuable sources of such data: they provide expert-authored problems with verified solutions, grounded in authentic pedagogical contexts across diverse disciplines. Early educational multimodal QA benchmarks such as TQA and ScienceQA already suggest that textbook-style materials are highly valuable sources of reasoning supervision [13, 19]. However, automatically extracting structured QA and VQA pairs from textbooks remains a challenging and largely underexplored problem. While modern OCR-based document parsing systems can reliably recover low-level layout elements, such as text blocks, equations, tables, and figures, the resulting outputs lack the semantic structure required for downstream learning[11, 15, 20, 23, 29]. Questions and answers are often separated by dozens of pages, figures are detached from the problems that reference them, and multi-part exercises are fragmented across columns and page boundaries. Without resolving these long-range logical dependencies, the parsed content cannot be directly transformed into usable training data.

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Conference acronym 'XX, Woodstock, NY
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ACM ISBN 978-1-4503-XXXX-X/2018/06
https://doi.org/XXXXXXX.XXXXXXX

Existing alternatives to textbook-based data construction suffer from complementary limitations. **Synthetic data generation** pipelines can efficiently produce large volumes of structured samples , and self-instruct style expansion has greatly lowered the marginal cost of dataset creation [25, 26]; however, such data often lack authentic context, compositional depth, and pedagogical diversity, limiting their effectiveness for real-world reasoning. In contrast, **expert-curated educational datasets** better reflect realistic learning scenarios [13, 19]; however, their reliance on domain experts for annotation, verification, and quality control makes them difficult to scale, both in terms of cost and time, thereby constraining dataset size and diversity.

To address these challenges, we introduce **FlipVQA-Miner**, an automated pipeline that extracts high-quality QA and VQA pairs directly from OCR-parsed pedagogical documents. Unlike prior document parsing approaches that primarily focus on layout reconstruction[11, 15, 20, 20, 34], our framework introduces an Identifier-Guided Semantic Reconstruction mechanism . This design combines layout-aware parsing with LLM-assisted block grouping and logical alignment, enabling the recovery of coherent question-answer-figure associations even when answers are located far from their corresponding questions, such as at the end of a book or in companion solution manuals.

Building upon this extraction framework, we further develop a multi-stage data curation pipeline that transforms raw extractions into training-ready supervision signals through structured question reconstruction, answer verification, question-type classification, and quality filtering. By replacing traditional expert-driven annotation with automated reconstruction and curation, FlipVQA-Miner reduces dataset construction cost by over **50**× while maintaining high data fidelity.

Using this system, we construct **FlipVQA-85K**, a large-scale reasoning dataset containing 85K QA and VQA instances spanning 11 academic disciplines. Extensive experiments demonstrate the effectiveness of the proposed data. Fine-tuning Qwen3-8B-Base [30] on FlipVQA consistently improves reasoning performance across multiple subject benchmarks, surpassing the stronger instruction-tuned baseline Qwen3-8B-Instruct [30]. In multimodal evaluation, fine-tuning Qwen3-VL-8B-Instruct [1] on FlipVQA also shows strong improvement on multiple VQA benchmarks, highlighting the value of high-quality textbook-derived supervision.

Contributions. (1) Automated Pedagogical Data Mining. We introduce FlipVQA-Miner, an automated pipeline for mining human-authored educational materials, enabling scalable construction of high-fidelity QA and VQA data with minimal hallucination.

(2) A Large-Scale Multi-Disciplinary Reasoning Dataset. We build FlipVQA-85K, a dataset consisting of 85K QA and VQA instances across 11 academic disciplines, derived directly from authentic pedagogical sources.

(3) Cost-Efficient Data Construction. Compared with expert-driven annotation pipelines, FlipVQA-Miner reduces data construction cost by over **50**×, significantly improving scalability without sacrificing data quality.

(4) Strong Empirical Performance. Fine-tuning Qwen3-8B-Base and Qwen3-VL-8B-Instruct on FlipVQA-85K yield consistent gains across multiple reasoning benchmarks, surpassing Qwen3-8B-Instruct and outperforming Qwen3-VL-8B-Instruct on VQA tasks.

2 Related Work

2.1 Document Intelligence and Semantic Reconstruction.

Document intelligence systems have transitioned from basic OCR to sophisticated layout-aware parsing. Traditional tools like Tesseract convert scanned pages into text but often discard crucial spatial metadata [23]. Modern frameworks, including LayoutLMv3 , DocLayout-YOLO , Pix2Struct , and MinerU 2.5 , have achieved impressive accuracy in recovering structural elements such as tables, equations, and multi-column layouts[2, 11, 14, 15, 20, 29, 34]. However, these approaches are primarily optimized for document readability and information extraction rather than the synthesis of structured supervision signals, which do not address the semantic alignment required for training reasoning-oriented models, such as identifying question-answer boundaries or linking figures to dispersed textual context.

2.2 Multimodal Reasoning and Instruction Datasets.

The evolution of Multimodal Large Language Models (MLLMs) has been significantly propelled by instruction-tuning datasets such as LLaVA-Instruct-150K, M3IT, ScienceQA, and TQA [13, 16, 17, 19]. These datasets have established critical benchmarks for multimodal alignment and reasoning evaluation. However, existing corpora predominantly rely on synthetic generation via self-instruct pipelines or web-scale image-text pairing, which often introduces hallucination, stylistic homogeneity, and limited generalization. Furthermore, while benchmarks like MathVista address scientific reasoning [8, 18, 31], they are typically manually curated at a small scale or lack an automated pipeline to transform raw pedagogical materials into structurally coherent supervision. Specifically, existing datasets fail to preserve long-range, cross-page logical dependencies inherent in educational documents.

3 Methodology

In this section, we introduce **FlipVQA-Miner** for high-fidelity knowledge extraction, and **Multi-stage Data Curation Pipeline** to curate the extracted data ready for training.

3.1 FlipVQA-Miner: Multimodal Knowledge Extraction

The extraction phase employs a hybrid design that integrates layout-aware document parsing with LLM-based semantic reconstruction. Crucially, our approach is modality-agnostic, enabling the synthesis of both text-only Question-Answer (QA) pairs and multimodal Visual Question-Answer (VQA) pairs.

3.1.1 Structural Content Parsing. We first utilize MinerU 2.5 [20], a powerful VLM-based OCR tool, to convert raw PDF files into a structured, machine-readable format. This step decomposes the document into fine-grained content blocks including text, equations, and embedded images while preserving their spatial metadata and layout information. By establishing this structural foundation, we provide a clean decomposition that reduces the complexity for subsequent semantic processing.

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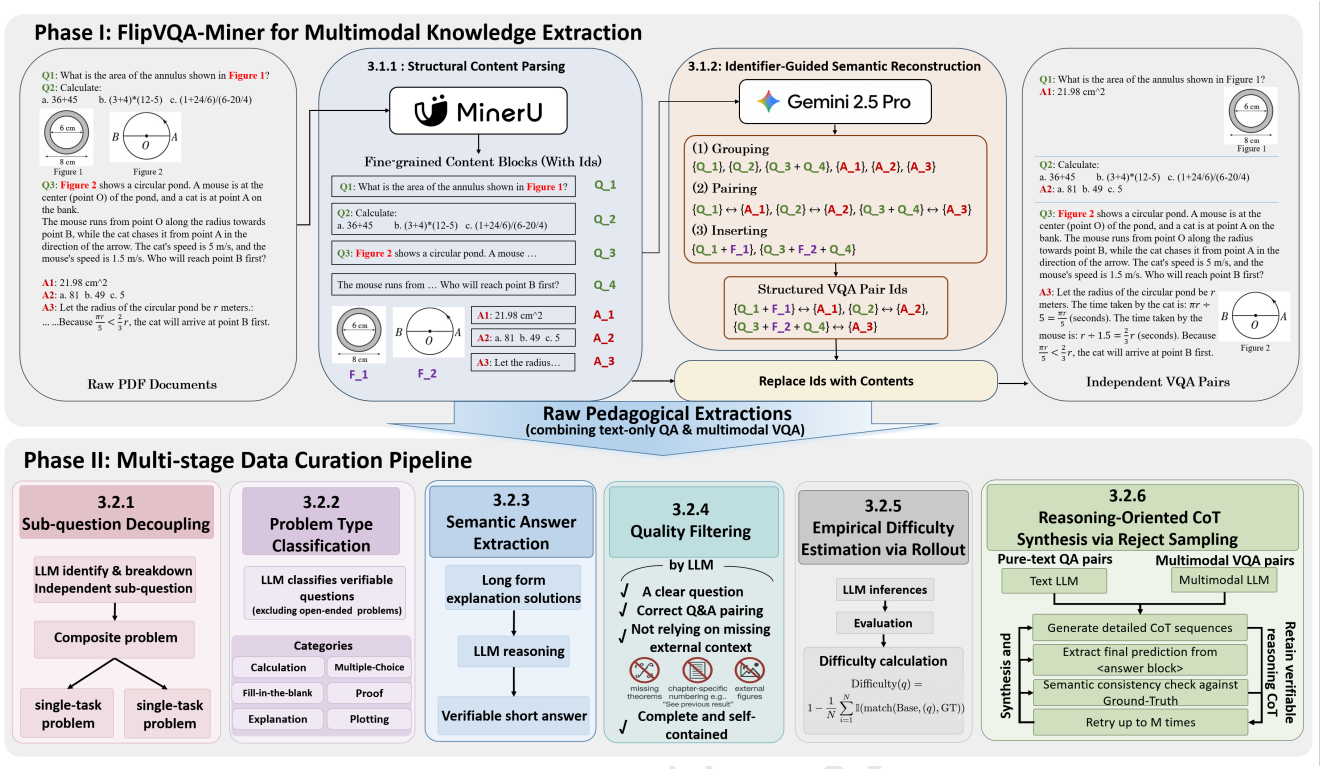

Figure 1: Overview of FlipVQA: The pipeline consists of two primary stages: (1) FlipVQA-Miner, which extract structured QA and VQA from Textbooks; and (2) a Multi-stage Data Curation Pipeline, which curate the extracted data ready for training.

3.1.2 Identifier-Guided Semantic Reconstruction (ISR). To resolve long-range dependencies and semantic discontinuities, the LLM operates on unique block identifiers rather than raw content, substantially reducing the number of *output tokens*. The system adaptively handles two extraction scenarios (see the prompt in section B.1): for **text-only QA extraction**, where sections are devoid of visual aids, the LLM focuses on **grouping** fragmented text and **pairing** questions with their corresponding answers across pages; for **multimodal VQA synthesis**, where visual evidence is present, the system additionally invokes an **inserting** operation, precisely placing relevant images into their associated textual context to form coherent VQA units.

3.2 Multi-stage Data Curation Pipeline

To transform raw pedagogical extractions into high-fidelity, "training-ready" supervision signals, we implement a rigorous multi-stage curation pipeline. It ensures that the processed dataset is clean, verifiable, and self-contained, effectively bridging the gap between fragmented textbook content and structured instruction-tuning requirements. The LLM we use in this pipeline is GPT-5-mini [22].

3.2.1 Sub-question Decoupling. Pedagogical materials frequently contain compound questions comprising multiple sub-questions that focus on one task (e.g. computing definite integral) but with different numbers. To ensure that each data entry poses a singular reasoning task, we employ a Sub-question Decoupling mechanism. Utilizing an LLM-based template generator, we identify and break down independent sub-problems while maintaining their relevant contexts. We carefully design the prompt so that context-sensitive sub-questions are not extracted. The prompt is provided in section B.2.

3.2.2 Problem Type Classification. To facilitate objective evaluation, we perform Problem Type Classification on the extracted corpus. An LLM is prompted to categorize each problem into specific types, including *Calculation*, *Multiple-choice*, *Fill-in-the-blank*, *Proof*, *Explanation* or *Plotting*. In our dataset, we prioritize verifiable types (Calculation, Multiple-choice and Fill-in-the-blank) and exclude open-ended or proof-based problems that lacks objective verification metrics. The prompt is provided in section B.3.

3.2.3 Semantic Answer Extraction. Textbook solutions often provide exhaustive, long-form explanations rather than concise, verifiable answers. We implement an Answer Extraction Operator that leverages LLM reasoning to derive canonical short answers from these narrative solutions, enabling automated evaluation. The prompt is provided in section B.4.

3.2.4 Quality Filtering. To ensure each sample is pedagogically sound and self-contained, we apply a strict Quality Filtering protocol. An LLM is prompted to ensure that the question raises a clear problem (e.g., not stating a fact), and the question and solution are correctly paired. It also eliminate instances that rely on missing external context, such as references to prior theorems, chapter-specific numbering (e.g., "See previous result"), or external figures.

Each sample is thus verified to be complete and self-contained, providing all necessary information for the model to reach a solution independently. The prompt is provided in section B.5.

3.2.5 Empirical Difficulty Estimation via Rollout. We quantify problem difficulty using an Empirical Rollout Evaluation ($N = 32$). The difficulty coefficient is defined as $1 - \text{Avg@}N$ accuracy:

$$Difficulty(q) = 1 - \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(\text{match}(\text{Base}_i(q), \text{GT})),$$

where Base represents the base model before fine-tuning, GT is the groundtruth answer. This enables the selection of challenging yet solvable samples, maintaining a reasonable difficulty distribution for training.

3.2.6 Reasoning-Oriented CoT Synthesis via Reject Sampling. While textbooks provide high-quality problems and answers, their intermediate reasoning steps are often presented in a format designed for human pedagogy rather than machine supervision. To bridge this gap and endow models with explicit logical trajectories, we implement a Reject Sampling-based CoT Synthesis pipeline. Depending on the input modality, we utilize different reasoning-specialized models to generate detailed Chain-of-Thought (CoT) sequences ($\mathcal{T}$): specifically, Qwen3-235B-A22B-Thinking is employed for pure-text QA pairs, while Qwen3-VL-235B-A22B-Thinking is utilized for VQA pairs to ensure precise visual grounding. To ensure the faithfulness and logical integrity of the synthesized reasoning, we adopt a *multi-round verification protocol*: for each generated candidate, we extract the final prediction from the `<answer>` block and perform a semantic consistency check against the ground-truth answer (GT). Only those trajectories that successfully lead to the correct result are retained:

$$\mathcal{S} = \{(q, \mathcal{T}) \mid \text{Verify}(\text{Extract}(\mathcal{T}), \text{GT}) = \text{True}\}$$

For instances where the initial generation fails, the pipeline performs up to M ($M = 5$ in our work) retries to maximize data yield without compromising quality. This iterative rejection and refinement process ensures that the resulting FlipVQA-85K dataset provides not only correct answers but also robust, verifiable reasoning chains for multimodal instruction tuning.

4 FlipVQA-85k Dataset

4.1 Data Collection

To construct a high-fidelity reasoning benchmark, we curate a corpus of 544 college-level educational PDF documents, including expert-authored textbooks and exercise sets[1]. The collection spans 11 academic disciplines, primarily in STEM domains where problems typically involve rigorous and verifiable reasoning processes. The selected documents exhibit complex layouts, such as multi-column structures and cross-page dependencies, posing realistic challenges for multimodal understanding.

4.2 Dataset Statistics and Comparisons

Scale and Diversity. FlipVQA comprises 85,968 questions, including 13,842 VQA pairs and 72,126 QA pairs, which ensures sufficient coverage for fine-tuning large multimodal models. The dataset spans 11 disciplines, fundamental sciences (e.g., Mathematics, Physics) to applied engineering (e.g., Aerospace, Automotive).

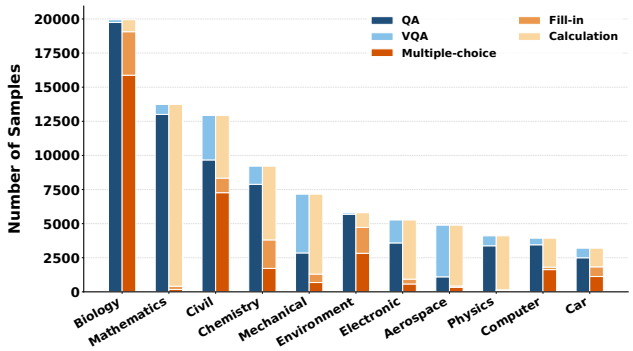

Figure 2: Question types and QA/VQA ratio for different subjects in our FlipVQA dataset.

Cross-Disciplinary Composition. We maintain multimodal coverage across all subjects, though the VQA-to-QA ratio varies by domain. Fig. 2 illustrates Mechanical Engineering and Aerospace exhibit a higher density of visual questions, whereas Biology remains predominantly textual. In terms of task types, FlipVQA is dominated by Calculation (52.5%), followed by Multiple-choice (35.8%) and Fill-in-the-blank (11.7%). The distribution of these tasks is highly subject-dependent; for instance, Calculation is prevalent in formal sciences, while Multiple-choice is more frequent in Biology and Civil Engineering, reflecting the distinct reasoning paradigms of each field.

5 Experiments and Analysis

5.1 Experiment Settings

Models. Our pipeline leverages a suite of specialized models for data processing and synthesis. For structural parsing, we employ MinerU 2.5 [20] to handle layout recognition and reading order. Gemini-2.5-pro [4] serves as the core LLM for logical QA pairing and image insertion. For reasoning-oriented CoT synthesis, we utilize Qwen3-235B-A22B-Thinking [30] (for text-only QA) and Qwen3-VL-235B-A22B-Thinking [1] (for VQA) to generate Chain-of-Thought (CoT) trajectories. We implement a rejection sampling protocol with up to $M = 5$ retries, retaining only those trajectories whose final predictions match the ground-truth answers to ensure data fidelity.

Benchmarks. We evaluate our method on two fronts: (1) **Extraction quality**, we select three representative documents with distinct layouts: a solution manual with immediate answers, an abstract algebra textbook with decoupled answers, and a multi-column Chinese workbook with appendix-based answers[2]. (2) **Downstream performance**, we evaluate on nine QA benchmarks and 6 VQA benchmarks, including general reasoning datasets (MMLU[9],

[1]All documents are sourced from open-access repositories under permissive licenses (e.g., CC-BY 4.0 or Public Domain), ensuring ethical reuse and benchmark transparency.

[2]Representative samples and visual layout demonstrations for each document are provided in Appendix A.

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Document	Type	Layout	Modality	#Samples	Precision	Recall	F1
Solutions Manual for Complex Analysis	Interleaved	Single-column	Text	390	0.9797	0.9897	0.9847
			Vision	294	1.0000	0.9456	0.9720
Textbook for Abstract Algebra	Long-distance	Single-column	Text	1281	0.9968	0.9766	0.9866
			Vision	27	1.0000	0.9259	0.9615
Exercise Book of Chinese Middle School Math	Long-distance	Multi-column	Text	445	0.9911	0.9955	0.9933
			Vision	266	1.0000	0.9774	0.9886

Table 1: Extraction performance of FlipVQA-Miner on different structural pattern documents. #Samples indicates the number of QA pairs for text and the number of images for vision. Representative samples and visual layout demonstrations for each document are provided in Appendix A.

MMLU-Pro[27], BBH[24], SuperGPQA[5], MMMU[31], OlympicArena [12]) and domain-specific reasoning benchmarks (GSM8K[3], MATH500[10], AIME2024, AIME2025, UGPhysics[28], MathVerse[33], MathVista[18], OlympiadBench[8], MatSciBench[32]). Detailed descriptions are provided in Appendix E.

Evaluation Protocols. **Extraction quality** is manually assessed across two modalities: (1) Textual Alignment, verifying the completeness and logical ordering of QA pairs; and (2) Visual Integrity, ensuring correct image placement. We focus on high-level structural errors rather than minor OCR noise, as MinerU 2.5 maintains a high textual F1 score above 0.94. Two human annotators were instructed to label the correctness of the extracted VQA pairs independently, and the inconsistent ones were verified by a third person. Detailed guidelines for annotators are provided in Appendix D.

Downstream performance evaluations employ greedy decoding (Temperature=0). Due to response complexity, we adopt an LLM-as-judge protocol using GPT-5-mini [22], following the HLE [6] setup. For the low-sample AIME benchmarks, we report the average of 32 rollouts per question to ensure statistical stability. Prompt details are provided in Appendix C.2.

Training Protocol. We perform supervised fine-tuning (SFT) on Qwen3-8B-Base [30] (for QA) and Qwen3-VL-8B-Instruct [1] (for VQA). For multimodal tuning, we freeze the vision encoder and update only the projector and language backbone. All models are trained using the LLaMA-Factory [35] framework with DeepSpeed ZeRO-3 [21] acceleration. Hyperparameters are detailed in section C.1.

5.2 Performance of FlipVQA-Miner

5.2.1 Quantitative Performance. Table 1 presents the quantitative results for both text and vision modalities on the documents. The pipeline achieves text F1 scores above 0.98 across all documents. Image placement precision is consistently 1.0, with F1 scores above 0.96. These results indicate that our pipeline achieves consistently high extraction accuracy for interleaved sequences, long-distance QA pairs, multi-column layouts, and multilingual content, demonstrating its robustness and potential for broader applications in automated VQA pair extraction.

To further demonstrate the qualitative results in resolving complex structural ambiguities, we provide a series of representative case studies in Appendix A.

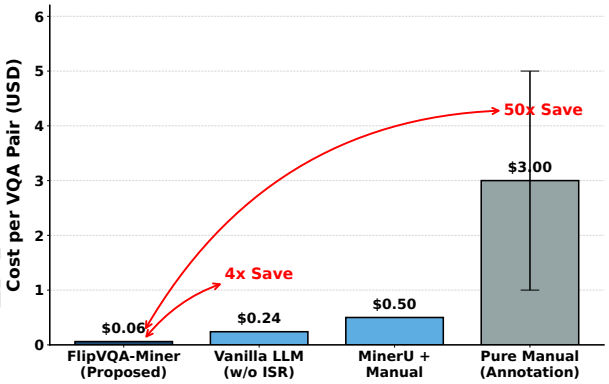

Figure 3: Cost-efficiency analysis of our VQA data extraction paradigms.

5.2.2 Cost-Efficiency. Beyond high extraction accuracy, another critical advantage of FlipVQA-Miner lies in its operational economy. Figure 3 illustrates a cost-efficiency analysis of FlipVQA-Miner against alternative VQA data synthesis paradigms. By leveraging the Identifier-guided Semantic Reconstruction (ISR) strategy (Section 3.1.2), our pipeline significantly mitigates computational and operational overhead. Specifically, transitioning from holistic LLM-based VQA generation to ISR-driven synthesis reduces the unit cost from $0.24 to $0.06, a fourfold (4×) improvement in efficiency over vanilla LLM generation. Furthermore, FlipVQA-Miner proves to be approximately 8× more cost-effective than MinerU-assisted human workflows which cost approximately $0.50 each pair. Compared to traditional manual annotation, which cost $1.00 to $5.00 each pair, FlipVQA-Miner yields an average 50× reduction in expenditure, highlighting its economic scalability for the curation of large-scale multimodal reasoning benchmarks.

5.3 Performance of Various Models Trained on FlipVQA-85K Dataset

The main results are summarized in Table 2. Overall, our method consistently outperforms the baselines across both textual and multimodal reasoning benchmarks.

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Table 2: Main results on reasoning benchmarks with average scores. Top: Results on textual benchmarks using Qwen3-8B-Base as the base model. Bottom: Results on multimodal benchmarks using Qwen3-VL-8B-Instruct as the base model. Bold indicates the best performance.

Textual Reasoning Benchmarks (QA)

	General				Math					Physics	
Model	MMLU	MMLU-Pro	BBH	SuperGPQA	GSM8K	MATH500	AIME24	AIME25	AIME26	UGPhys	**Average Acc.**
Qwen3-8B-Base	65.26	47.16	68.55	24.21	77.63	67.80	14.27	14.69	12.08	22.61	41.41
Qwen3-8B (Non-thinking)	78.78	65.39	80.99	35.90	93.40	84.80	25.31	**21.25**	16.46	36.18	53.85
Ours (Qwen3-8B-Base-sft)	**81.12**	**67.45**	**81.37**	**37.88**	**95.00**	**87.40**	**25.73**	20.31	**18.33**	**39.95**	**55.45**

Multimodal Reasoning Benchmarks (VQA)

	General			Math+Physics	Math		Chemistry	
Model	FlipVQA-test	MMMU	OlympicArena	OlympiadBench	MathVerse	MathVista	MatSciBench	**Average Acc.**
Qwen3-VL-8B-Instruct	24.45	65.78	42.54	57.35	66.73	**70.30**	29.79	50.99
Ours	**32.97**	**66.22**	**47.39**	**64.14**	**71.37**	70.10	**35.62**	**55.40**

On textual benchmarks, our fine-tuned model improves performance on 9 out of 10 datasets compared with the Qwen3-8B Non-thinking mode. In particular, notable gains are observed on MMLU, Math500, and UGPhysics, indicating improvements in both general reasoning ability and domain knowledge in scientific subjects.

On multimodal benchmarks, our model achieves improvements on 5 out of 7 datasets compared with the Qwen3-VL-8B baseline, showing enhanced visual reasoning performance across disciplines such as mathematics, physics, and chemistry. On MMMU, the model exhibits a modest improvement, indicating that the specialization introduced by fine-tuning does not compromise its general-purpose multimodal capability.

A slight performance decrease is observed on MathVista, likely due to distributional differences between the training and evaluation data. Specifically, FlipVQA-85k mainly consists of problems extracted from college-level textbooks, whereas MathVista focuses more on logic puzzles, functional plots, and figures commonly found in academic publications.

5.4 Ablation Study: Impact of Data Curation

To quantify the impact of our **Multi-stage Data Curation Pipeline**, we compare the final FlipVQA model against a baseline trained on raw extractions from the FlipVQA-Miner. As shown in Table 3, our data curation significantly outperforms the raw extraction baseline. While textbook corpora provide high fidelity problem and answer pairs, enhancing latent reasoning requires meticulously curated long form CoT trajectories. Without multi-stage data curation, training on raw data lacking structured reasoning fails to surpass or may even degrade the inherent reasoning priors of Qwen3-VL-8B-Instruct. Thus, rigorous data curation and CoT synthesis are indispensable for distilling textbook knowledge into executable multimodal intelligence.

Table 3: Ablation study of the FlipVQA curation pipeline. Phase 1 refers to FlipVQA-Miner and Phase 2 represents the Multi-stage Data Curation Pipeline.

Textual Reasoning Benchmarks (QA)

Model	**Phase 1**	**Phase 2**	**MMLU-Pro**	**SuperGPQA**	**Avg.**
Qwen3-8B-Base	×	×	47.16	24.21	35.69
Qwen3-8B(Non-thinking)	×	×	65.39	35.90	50.65
Raw extraction only	✓	×	32.11	15.53	23.82
FlipVQA-85k(QA)	✓	✓	**67.11**	**37.39**	**52.25**

Multimodal Reasoning Benchmarks (VQA)

Model	**Phase 1**	**Phase 2**	**MMMU**	**OlympicArena**	**Avg.**
Qwen3-VL-8B-Instruct	×	×	65.78	42.54	54.16
Raw extraction only	✓	×	53.67	30.97	42.32
FlipVQA-85K(VQA)	✓	✓	**66.22**	**47.39**	**56.81**

6 Conclusion

We have presented **FlipVQA-Miner**, an automated pipeline that synthesizes high-fidelity, reasoning-oriented VQA pairs from multidisciplinary textbooks. By effectively resolving long-range logical dependencies and cross-page discontinuities inherent in pedagogical documents, our framework transforms raw content into grounded, hallucination-free supervision signals. We have demonstrated that FlipVQA-Miner achieves a **50**× cost saving compared to manual annotation while maintaining exceptional structural fidelity ($F_1 > 0.96$). Our multi-stage data curation pipeline is critical for transforming raw extractions into grounded supervision signals through curated long form Chain of Thought trajectories. Utilizing this pipeline, we released the **FlipVQA-85K**, comprising 85k instances spanning 11 academic disciplines. Models fine-tuned on FlipVQA-85K show significantly improved reasoning robustness, surpassing stronger instruction-tuned baselines across both textual and multimodal benchmarks.

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A Case Studies of VQA Pair Extraction

To demonstrate the robustness of our pipeline in resolving diverse structural ambiguities commonly found in educational materials, we present a set of representative examples. Each example includes the original question, the original solution(s), and the corresponding extracted QA or VQA pair rendered in Markdown. These cases illustrate how the system reconstructs raw, unstructured content into a coherent, machine-readable format.

A.1 Long-distance, cross-document VQA association case

This case demonstrates FlipVQA-Miner's capability to resolve extreme structural decoupling where components of a single VQA triplet are distributed across distant pages and even different materials. As illustrated in Figure 4, the question (①) refers to specific "square regions", yet the associated figure (②) is located on a subsequent page and lacks any explicit caption or label. Furthermore, the corresponding answer (③) is found in a separate companion volume rather than the primary document. This geographic separation poses a significant challenge for conventional extraction tools. Despite these hurdles, FlipVQA-Miner successfully identifies the latent semantic links, effectively reassembling the fragmented elements into a unified, structured VQA unit. The final Markdown-rendered output, showcasing the system's ability to maintain logical coherence across long-distance associations, is shown in Figure 5.

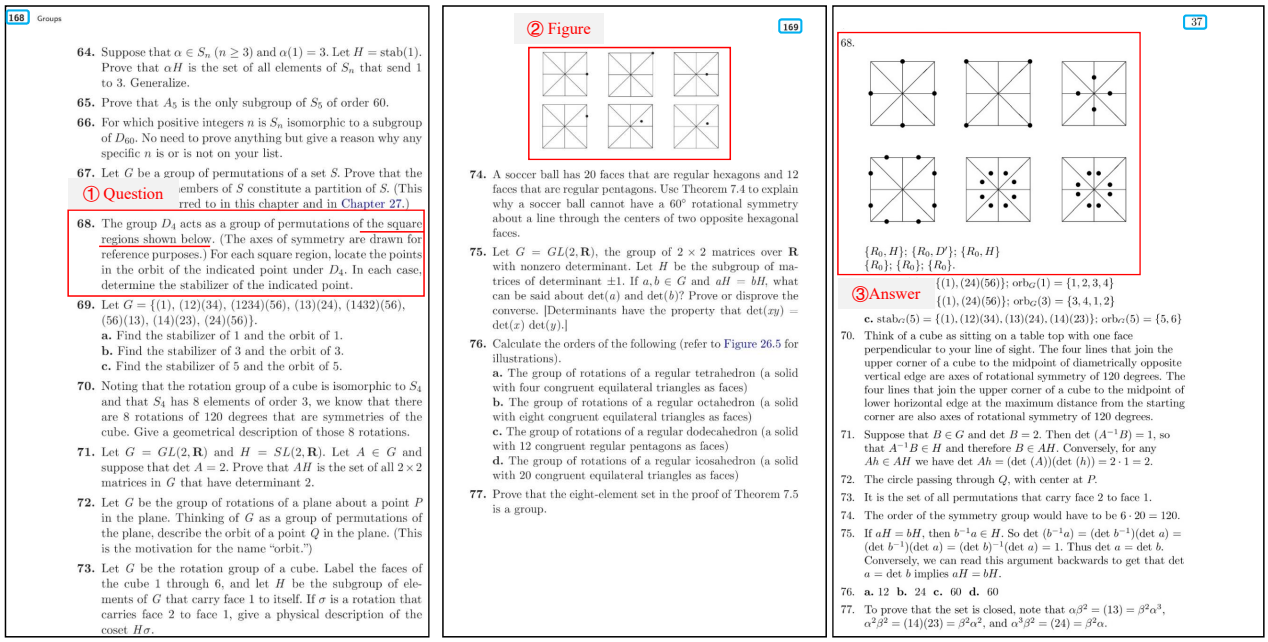

Figure 4: Original arrangement in PDF where the question, figure, and answer are separated across pages and documents.

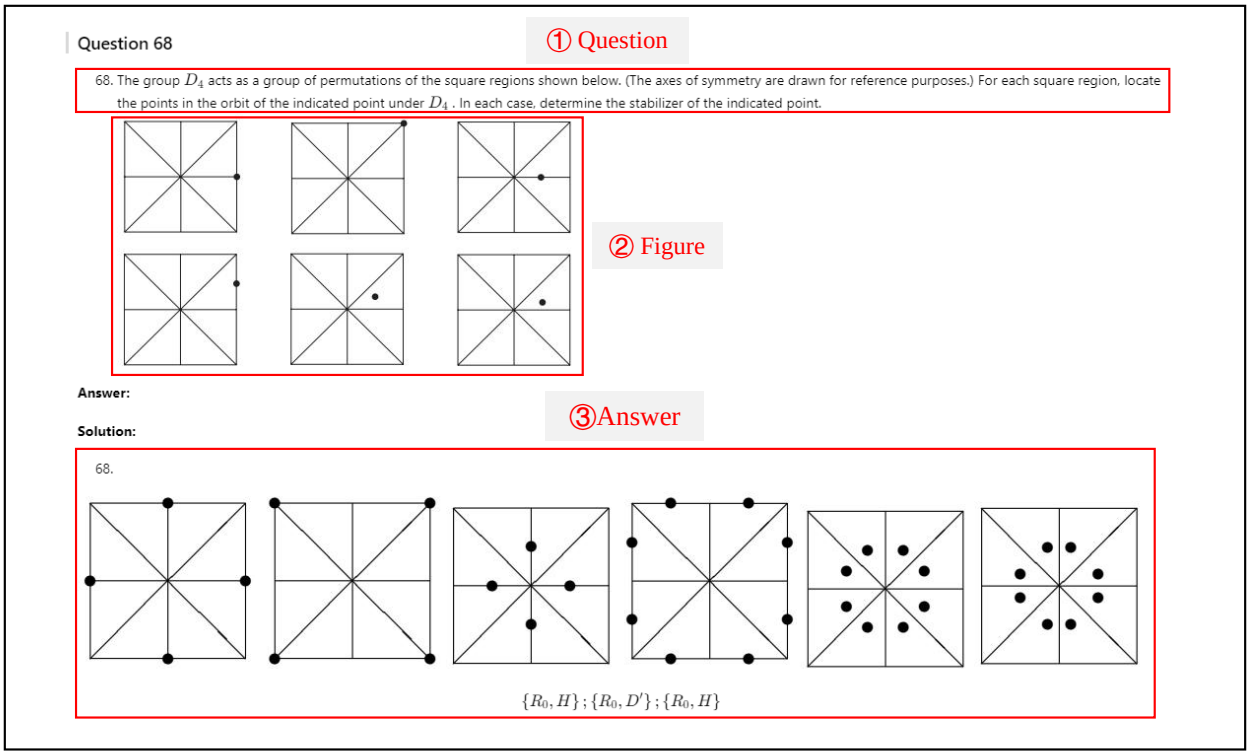

Figure 5: Extracted and reconstructed VQA pair rendered in Markdown, showing successful long-distance grouping.

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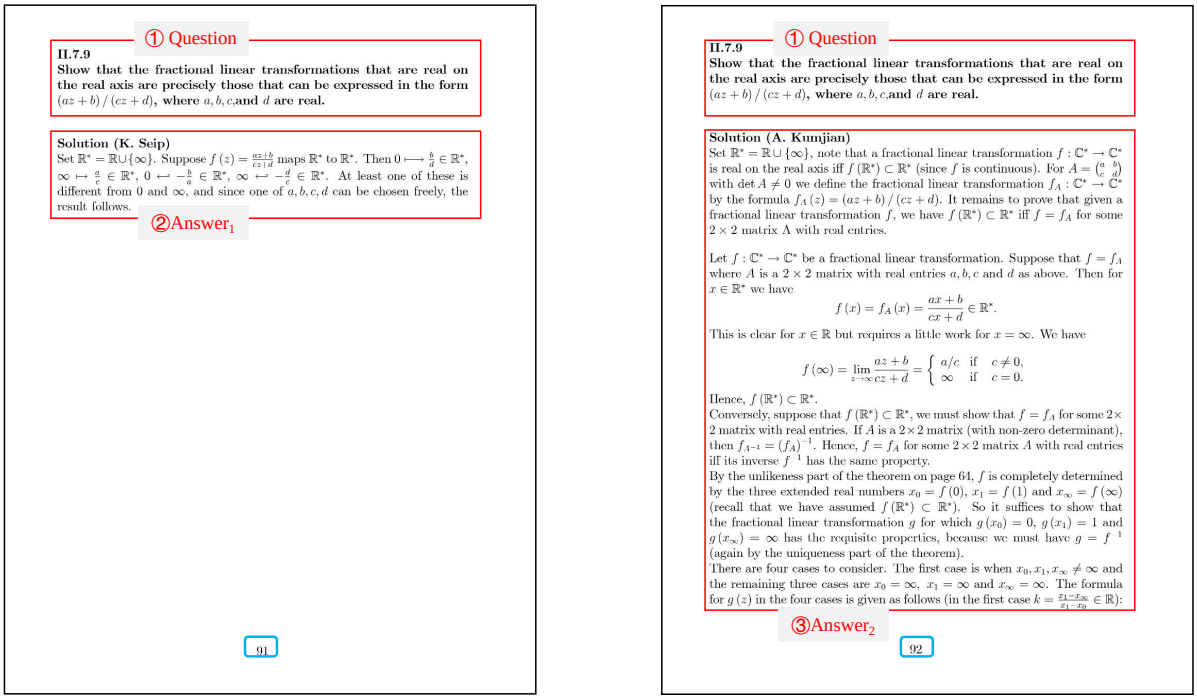
A.2 Interleaved QA with multiple solutions case

This case features an interleaved question (Figure 6) accompanied by two distinct solution paths. Despite the intertwined layout, our pipeline faithfully preserves both solutions and correctly associates them with the question. The resulting structured output is shown in Figure 7.



① Question

II.7.9
Show that the fractional linear transformations that are real on the real axis are precisely those that can be expressed in the form $(az+b)/(cz+d)$, where $a, b, c,$and d are real.

Solution (K. Seip)
Set $\mathbb{R}^* = \mathbb{R} \cup \{\infty\}$. Suppose $f(z) = \frac{az+b}{cz+d}$ maps $\mathbb{R}^*$ to $\mathbb{R}^*$. Then $0 \longmapsto \frac{b}{d} \in \mathbb{R}^*$, $\infty \mapsto \frac{a}{c} \in \mathbb{R}^*$, $0 \hookleftarrow -\frac{b}{a} \in \mathbb{R}^*$, $\infty \hookleftarrow -\frac{d}{c} \in \mathbb{R}^*$. At least one of these is different from 0 and ∞, and since one of a, b, c, d can be chosen freely, the result follows.

②Answer$_1$

91

① Question

II.7.9
Show that the fractional linear transformations that are real on the real axis are precisely those that can be expressed in the form $(az+b)/(cz+d)$, where $a, b, c,$and d are real.

Solution (A. Kumjian)
Set $\mathbb{R}^* = \mathbb{R} \cup \{\infty\}$, note that a fractional linear transformation $f : \mathbb{C}^* \to \mathbb{C}^*$ is real on the real axis iff $f(\mathbb{R}^*) \subset \mathbb{R}^*$ (since f is continuous). For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\det A \neq 0$ we define the fractional linear transformation $f_A : \mathbb{C}^* \to \mathbb{C}^*$ by the formula $f_A(z) = (az+b)/(cz+d)$. It remains to prove that given a fractional linear transformation f, we have $f(\mathbb{R}^*) \subset \mathbb{R}^*$ iff $f = f_A$ for some 2×2 matrix A with real entries.

Let $f : \mathbb{C}^* \to \mathbb{C}^*$ be a fractional linear transformation. Suppose that $f = f_A$ where A is a 2×2 matrix with real entries a, b, c and d as above. Then for $x \in \mathbb{R}^*$ we have

$$f(x) = f_A(x) = \frac{ax+b}{cx+d} \in \mathbb{R}^*.$$

This is clear for $x \in \mathbb{R}$ but requires a little work for $x = \infty$. We have

$$f(\infty) = \lim_{z \to \infty} \frac{az+b}{cz+d} = \begin{cases} a/c & \text{if} \quad c \neq 0, \\ \infty & \text{if} \quad c = 0. \end{cases}$$

Hence, $f(\mathbb{R}^*) \subset \mathbb{R}^*$.
Conversely, suppose that $f(\mathbb{R}^*) \subset \mathbb{R}^*$, we must show that $f = f_A$ for some 2×2 matrix with real entries. If A is a 2×2 matrix (with non-zero determinant), then $f_{A^{-1}} = (f_A)^{-1}$. Hence, $f = f_A$ for some 2×2 matrix A with real entries iff its inverse f^{-1} has the same property.
By the unlikeness part of the theorem on page 64, f is completely determined by the three extended real numbers $x_0 = f(0)$, $x_1 = f(1)$ and $x_\infty = f(\infty)$ (recall that we have assumed $f(\mathbb{R}^*) \subset \mathbb{R}^*$). So it suffices to show that the fractional linear transformation g for which $g(x_0) = 0$, $g(x_1) = 1$ and $g(x_\infty) = \infty$ has the requisite properties, because we must have $g = f^{-1}$ (again by the uniqueness part of the theorem).
There are four cases to consider. The first case is when $x_0, x_1, x_\infty \neq \infty$ and the remaining three cases are $x_0 = \infty$, $x_1 = \infty$ and $x_\infty = \infty$. The formula for $g(z)$ in the four cases is given as follows (in the first case $k = \frac{x_1 - x_\infty}{x_1 - x_0} \in \mathbb{R}$):

③Answer$_2$

92

Figure 6: Original question with two interleaved solutions.

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Question 9

① Question

Show that the fractional linear transformations that are real on the real axis are precisely those that can be expressed in the form $(az+b)/(cz+d)$, where $a, b, c,$ and d are real.

Answer:

Solution:

②Answer$_1$

Solution (K. Seip) Set $\mathbb{R}^* = \mathbb{R} \cup \{\infty\}$. Suppose $f(z) = \frac{az+b}{cz+d}$ maps $\mathbb{R}^*$ to $\mathbb{R}^*$. Then $0 \longmapsto \frac{b}{d} \in \mathbb{R}^*$, $\infty \mapsto \frac{a}{c} \in \mathbb{R}^*$, $0 \longleftarrow -\frac{b}{a} \in \mathbb{R}^*$, $\infty \longleftarrow -\frac{d}{c} \in \mathbb{R}^*$. At least one of these is different from 0 and ∞ , and since one of a, b, c, d can be chosen freely, the result follows.

Solution (A. Kumjian) Set $\mathbb{R}^* = \mathbb{R} \cup \{\infty\}$, note that a fractional linear transformation $f : \mathbb{C}^* \to \mathbb{C}^*$ is real on the real axis iff $f(\mathbb{R}^*) \subset \mathbb{R}^*$ (since f is continuous). For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\det A \neq 0$ we define the fractional linear transformation $f_A : \mathbb{C}^* \to \mathbb{C}^*$ by the formula $f_A(z) = (az+b)/(cz+d)$. It remains to prove that given a fractional linear transformation f , we have $f(\mathbb{R}^*) \subset \mathbb{R}^*$ iff $f = f_A$ for some 2×2 matrix A with real entries. Let $f : \mathbb{C}^* \to \mathbb{C}^*$ be a fractional linear transformation. Suppose that $f = f_A$ where A is a 2×2 matrix with real entries a, b, c and d as above. Then for $x \in \mathbb{R}^*$ we have

$$f(x) = f_A(x) = \frac{ax+b}{cx+d} \in \mathbb{R}^*.$$

This is clear for $x \in \mathbb{R}$ but requires a little work for $x = \infty$. We have

$$f(\infty) = \lim_{z \to \infty} \frac{az+b}{cz+d} = \begin{cases} a/c & \text{if} \quad c \neq 0, \\ \infty & \text{if} \quad c = 0. \end{cases}$$

Hence, $f(\mathbb{R}^*) \subset \mathbb{R}^*$ Conversely, suppose that $f(\mathbb{R}^*) \subset \mathbb{R}^*$, we must show that $f = f_A$ for some 2×2 matrix with real entries. If A is a 2×2 matrix (with non-zero determinant), then $f_{A^{-1}} = (f_A)^{-1}$. Hence, $f = f_A$ for some 2×2 matrix A with real entries iff its inverse f^{-1} has the same property. By the unlikeness part of the theorem on page 64, f is completely determined by the three extended real numbers $x_0 = f(0)$, $x_1 = f(1)$ and $x_\infty = f(\infty)$ (recall that we have assumed $f(\mathbb{R}^*) \subset \mathbb{R}^*$). So it suffices to show that the fractional linear transformation g for which $g(x_0) = 0$, $g(x_1) = 1$ and $g(x_\infty) = \infty$ has the requisite properties, because we must have $g = f^{-1}$ (again by the uniqueness part of the theorem). There are four cases to consider. The first case is when $x_0, x_1, x_\infty \neq \infty$ and the remaining three cases are $x_0 = \infty$, $x_1 = \infty$ and $x_\infty = \infty$. The formula for $g(z)$ in the four cases is given as follows (in the first case $k = \frac{x_1 - x_\infty}{x_1 - x_0} \in \mathbb{R}$):

$$k\frac{x - x_0}{x - x_\infty} \quad \text{if} \quad x_0, x_1, x_\infty \neq \infty, \qquad \frac{x_1 - x_\infty}{x - x_\infty}\frac{x - x_0}{x_1 - x_0} \quad \text{if} \quad x_0 = \infty,$$
$$\frac{x - x_0}{x - x_\infty} \quad \text{if} \quad x_1 = \infty, \qquad \frac{x_1 - x_\infty}{x_1 - x_0} \quad \text{if} \quad x_\infty = \infty$$

In each case $g(z)$ has the desired form. Hence, $f = f_A$ for some 2×2 matrix A with real entries as required.

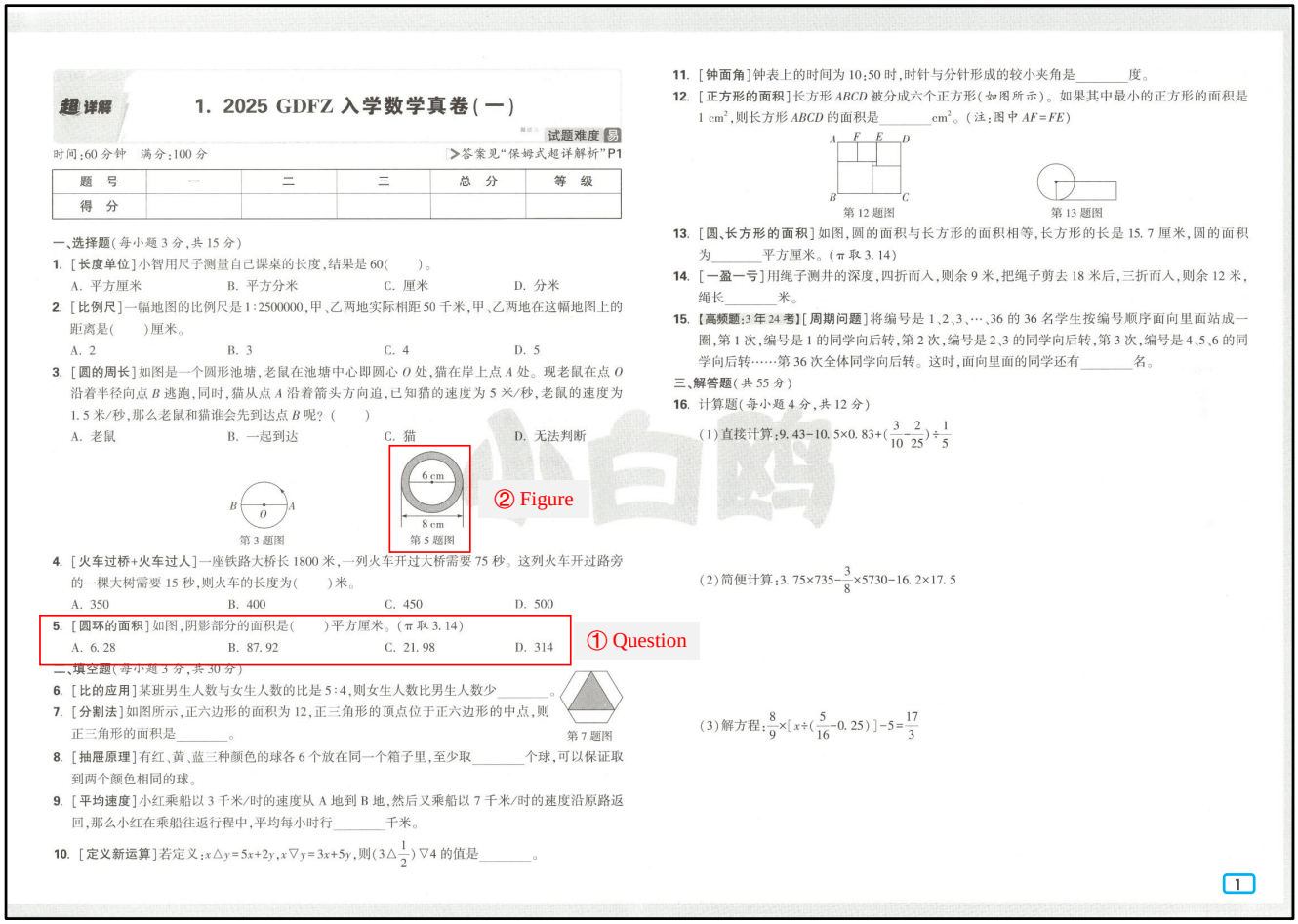
③Answer$_2$

Figure 7: Extracted and reconstructed QA pair rendered in Markdown.

Unpublished working draft. Not for distribution.

A.3 Multi-column, cross-page Chinese exercise book case

This example showcases a particularly challenging scenario drawn from a Chinese exercise book. As shown in Figure 8, the question appears within a dense multi-column layout interspersed with unrelated textual elements. Complicating matters further, the corresponding answer (Figure 9) is located in a separate PDF file altogether. Despite these structural and formatting hurdles, our pipeline accurately identifies, extracts, and associates the VQA pair. The final Markdown-rendered output (Figure 10) highlights the system's resilience to multi-column layouts, cross-page and cross-document separation, heterogeneous formatting conventions, and multilingual content.



超详解 1. 2025 GDFZ 入学数学真卷(一)

试题难度 易

时间:60 分钟 满分:100 分 >答案见"保姆式超详解析"P1

题 号	一	二	三	总 分	等 级
得 分					

一、选择题(每小题 3 分,共 15 分)

1. [长度单位]小智用尺子测量自己课桌的长度,结果是 60(　　)。
A. 平方厘米　B. 平方分米　C. 厘米　D. 分米

2. [比例尺]一幅地图的比例尺是 1∶2500000,甲、乙两地实际相距 50 千米,甲、乙两地在这幅地图上的距离是(　　)厘米。
A. 2　B. 3　C. 4　D. 5

3. [圆的周长]如图是一个圆形池塘,老鼠在池塘中心即圆心 O 处,猫在岸上点 A 处。现老鼠在点 O 沿着半径向点 B 逃跑,同时,猫从点 A 沿着箭头方向追,已知猫的速度为 5 米/秒,老鼠的速度为 1.5 米/秒,那么老鼠和猫谁会先到达点 B 呢?(　　)
A. 老鼠　B. 一起到达　C. 猫　D. 无法判断

第 3 题图

第 5 题图

② Figure

4. [火车过桥+火车过人]一座铁路大桥长 1800 米,一列火车开过大桥需要 75 秒。这列火车开过路旁的一棵大树需要 15 秒,则火车的长度为(　　)米。
A. 350　B. 400　C. 450　D. 500

5. [圆环的面积]如图,阴影部分的面积是(　　)平方厘米。(π 取 3.14)
A. 6.28　B. 87.92　C. 21.98　D. 314

① Question

二、填空题(每小题 3 分,共 30 分)

6. [比的应用]某班男生人数与女生人数的比是 5∶4,则女生人数比男生人数少_____。

7. [分割法]如图所示,正六边形的面积为 12,正三角形的顶点位于正六边形的中点,则正三角形的面积是_____。

第 7 题图

8. [抽屉原理]有红、黄、蓝三种颜色的球各 6 个放在同一个箱子里,至少取_____个球,可以保证取到两个颜色相同的球。

9. [平均速度]小红乘船以 3 千米/时的速度从 A 地到 B 地,然后又乘船以 7 千米/时的速度沿原路返回,那么小红在乘船往返行程中,平均每小时行_____千米。

10. [定义新运算]若定义:$x\triangle y=5x+2y$,$x\triangledown y=3x+5y$,则$(3\triangle\frac{1}{2})\triangledown 4$ 的值是_____。

11. [钟面角]钟表上的时间为 10:50 时,时针与分针形成的较小夹角是_____度。

12. [正方形的面积]长方形 $ABCD$ 被分成六个正方形(如图所示)。如果其中最小的正方形的面积是 1 cm^2,则长方形 $ABCD$ 的面积是_____cm^2。(注:图中 $AF=FE$)

第 12 题图

第 13 题图

13. [圆、长方形的面积]如图,圆的面积与长方形的面积相等,长方形的长是 15.7 厘米,圆的面积为_____平方厘米。(π 取 3.14)

14. [一盈一亏]用绳子测井的深度,四折而入,则余 9 米,把绳子剪去 18 米后,三折而入,则余 12 米,绳长_____米。

15. [高频题:3 年 24 考][周期问题]将编号是 1、2、3、…、36 的 36 名学生按编号顺序面向里面站成一圈,第 1 次,编号是 1 的同学向后转,第 2 次,编号是 2、3 的同学向后转,第 3 次,编号是 4、5、6 的同学向后转……第 36 次全体同学向后转。这时,面向里面的同学还有_____名。

三、解答题(共 55 分)

16. 计算题(每小题 4 分,共 12 分)

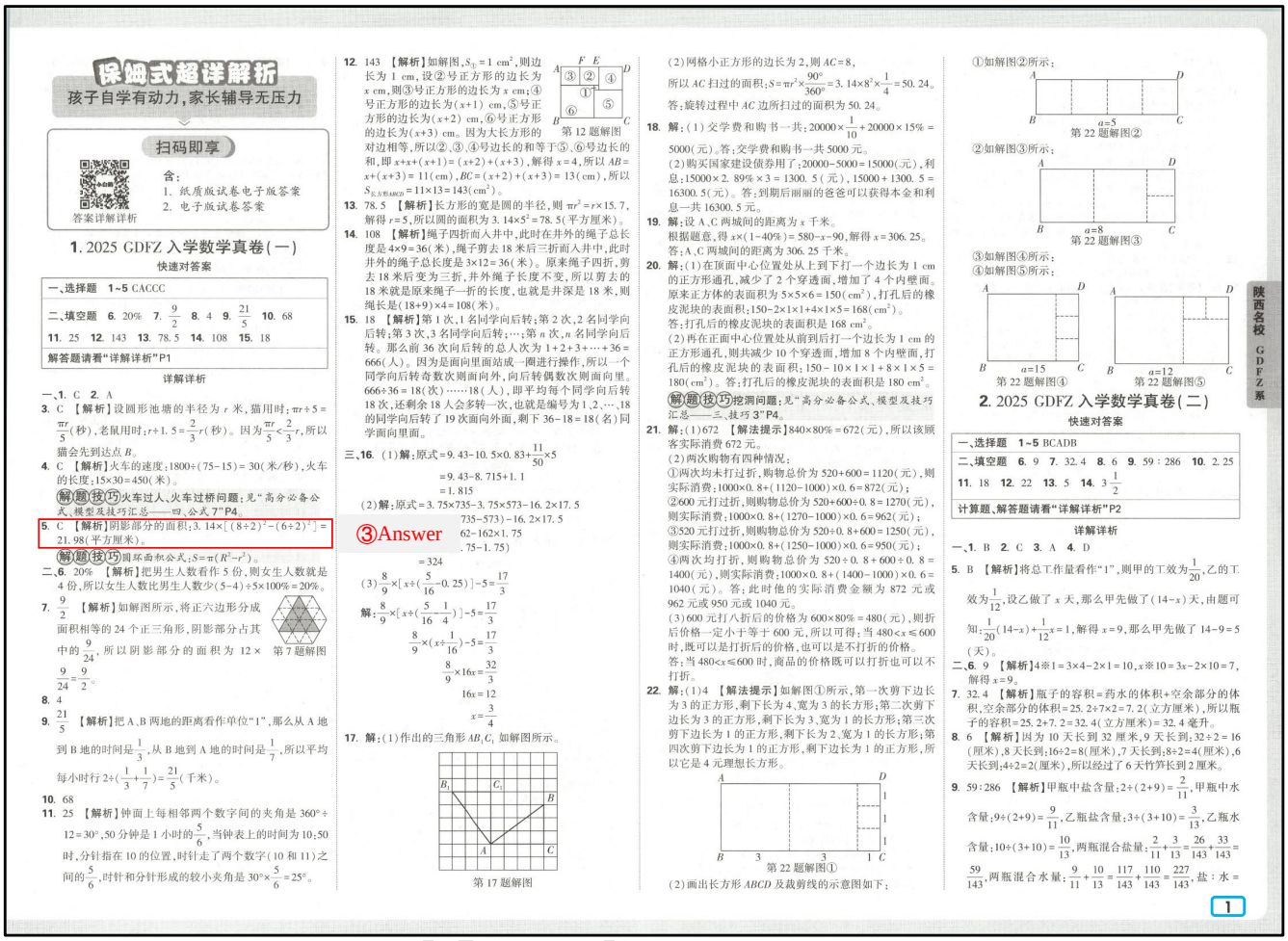
(1)直接计算:$9.43-10.5\times0.83+(\frac{3}{10}-\frac{2}{25})\div\frac{1}{5}$

(2)简便计算:$3.75\times735-\frac{3}{8}\times5730-16.2\times17.5$

(3)解方程:$\frac{8}{9}\times[x\div(\frac{5}{16}-0.25)]-5=\frac{17}{3}$

1

Figure 8: Original question within a multi-column Chinese exercise book layout.



保姆式超详解析

孩子自学有动力，家长辅导无压力

扫码即享

含：
1. 纸质版试卷电子版答案
2. 电子版试卷答案

答案详解详析

1. 2025 GDFZ 入学数学真卷(一)

快速对答案

一、选择题 1~5 CACCC	
二、填空题 6. 20% 7. $\frac{9}{2}$ 8. 4 9. $\frac{21}{5}$ 10. 68	
11. 25 12. 143 13. 78. 5 14. 108 15. 18	
解答题请看"详解详析"P1	

详解详析

一、1. C 2. A

3. C 【解析】设圆形池塘的半径为 r 米，猫用时：$\pi r\div 5=\frac{\pi r}{5}$(秒)，老鼠用时：$r\div 1.5=\frac{2}{3}r$(秒)。因为 $\frac{\pi r}{5}<\frac{2}{3}r$，所以猫会先到达点 B。

4. C 【解析】火车的速度：1800÷(75−15)=30(米/秒)，火车的长度：15×30=450(米)。

解题技巧 火车过人、火车过桥问题：见"高分必备公式、模型及技巧汇总——四、公式 7"P4。

5. C 【解析】阴影部分的面积：$3.14\times[(8\div 2)^2-(6\div 2)^2]=21.98$(平方厘米)。

解题技巧 圆环面积公式：$S=\pi(R^2-r^2)$。

二、6. 20% 【解析】把男生人数看作 5 份，则女生人数就是 4 份，所以女生人数比男生人数少(5−4)÷5×100%=20%。

7. $\frac{9}{2}$ 【解析】如解图所示，将正六边形分成面积相等的 24 个正三角形，阴影部分占其中的 $\frac{9}{24}$，所以阴影部分的面积为 $12\times\frac{9}{24}=\frac{9}{2}$。

第 7 题解图

8. 4

9. $\frac{21}{5}$ 【解析】把 A、B 两地的距离看作单位"1"，那么从 A 地到 B 地的时间是 $\frac{1}{3}$，从 B 地到 A 地的时间是 $\frac{1}{7}$，所以平均每小时行 $2\div(\frac{1}{3}+\frac{1}{7})=\frac{21}{5}$(千米)。

10. 68

11. 25 【解析】钟面上每相邻两个数字间的夹角是 360°÷12=30°，50 分钟是 1 小时的 $\frac{5}{6}$，当钟表上的时间为 10:50 时，分针指在 10 的位置，时针走了两个数字(10 和 11)之间的 $\frac{5}{6}$，时针和分针形成的较小夹角是 $30°\times\frac{5}{6}=25°$。

12. 143 【解析】如解图，$S_{①}=1\ cm^2$，则边长为 1 cm，设②号正方形的边长为 x cm，则③号正方形的边长为 x cm；④号正方形的边长为 $(x+1)$ cm，⑤号正方形的边长为 $(x+2)$ cm，⑥号正方形的边长为 $(x+3)$ cm。因为大长方形的对边相等，所以②、③、④号边长的和等于⑤、⑥号边长的和，即 $x+x+(x+1)=(x+2)+(x+3)$，解得 $x=4$，所以 $AB=x+(x+3)=11$(cm)，$BC=(x+2)+(x+3)=13$(cm)，所以 $S_{长方形ABCD}=11\times 13=143(cm^2)$。

第 12 题解图

13. 78. 5 【解析】长方形的宽是圆的半径，则 $\pi r^2=r\times 15.7$，解得 $r=5$，所以圆的面积为 $3.14\times 5^2=78.5$(平方厘米)。

14. 108 【解析】绳子四折面入井中，此时在井外的绳子总长度是 4×9=36(米)，绳子剪去 18 米后三折面入井中，此时井外的绳子总长度是 3×12=36(米)。原来绳子四折，剪去 18 米后变为三折，井外绳子长度不变，所以剪去的 18 米就是原来绳子一折的长度，也就是井深是 18 米，则绳长是(18+9)×4=108(米)。

15. 18 【解析】第 1 次，1 名同学向后转；第 2 次，2 名同学向后转；第 3 次，3 名同学向后转；…；第 n 次，n 名同学向后转。那么前 36 次向后转的总人次为 $1+2+3+\cdots+36=666$(人)。因为是面向里面站成一圈进行操作，所以一个同学向后转奇数次则面向外，向后转偶数次则面向里。666÷36=18(次)……18(人)，即平均每个同学向后转 18 次，还剩余 18 人会多转一次，也就是编号为 1，2，…，18 的同学向后转了 19 次面向外面，剩下 36−18=18(名)同学面向里面。

三、16. (1)解：原式 $=9.43-10.5\times 0.83+\frac{11}{50}\times 5$

$=9.43-8.715+1.1$

$=1.815$

(2)解：原式 $=3.75\times 735-3.75\times 573-16.2\times 17.5$

$735-573)-16.2\times 17.5$

$62-162\times 1.75$

$.75-1.75)$

$=324$

(3) $\frac{8}{9}\times[x\div(\frac{5}{16}-0.25)]-5=\frac{17}{3}$

解：$\frac{8}{9}\times[x\div(\frac{5}{16}-\frac{1}{4})]-5=\frac{17}{3}$

$\frac{8}{9}\times(x\div\frac{1}{16})-5=\frac{17}{3}$

$\frac{8}{9}\times 16x=\frac{32}{3}$

$16x=12$

$x=\frac{3}{4}$

17. 解：(1)作出的三角形 AB_1C_1 如解图所示。

第 17 题解图

(2)网格小正方形的边长为 2，则 $AC=8$，

所以 AC 扫过的面积：$S=\pi r^2\times\frac{90°}{360°}=3.14\times 8^2\times\frac{1}{4}=50.24$。

答：旋转过程中 AC 边所扫过的面积为 50. 24。

18. 解：(1)交学费和购书一共：$20000\times\frac{1}{10}+20000\times 15\%=5000$(元)。答：交学费和购书一共 5000 元。

(2)购买国家建设债券用了：20000−5000=15000(元)，利息：15000×2. 89%×3=1300. 5(元)，15000+1300. 5=16300. 5(元)。答：到期后丽丽的爸爸可以获得本金和利息一共 16300. 5 元。

19. 解：设 A、C 两城间的距离为 x 千米。

根据题意，得 $x\times(1-40\%)=580-x-90$，解得 $x=306.25$。

答：A、C 两城间的距离为 306. 25 千米。

20. 解：(1)在顶面中心位置处从上到下打一个边长为 1 cm 的正方形通孔，减少了 2 个穿透面，增加了 4 个内壁面。原来正方体的表面积为 5×5×6=150(cm^2)，打孔后的橡皮泥块的表面积：150−2×1×1+4×1×5=168(cm^2)。

答：打孔后的橡皮泥块的表面积是 168 cm^2。

(2)再在正面中心位置处从前到后打一个边长为 1 cm 的正方形通孔，则共减少 10 个穿透面，增加 8 个内壁面，打孔后的橡皮泥块的表面积：150−10×1×1+8×1×5=180(cm^2)。答：打孔后的橡皮泥块的表面积是 180 cm^2。

解题技巧 挖洞问题：见"高分必备公式、模型及技巧汇总——三、技巧 3"P4。

21. 解：(1)672 【解法提示】840×80%=672(元)，所以该顾客实际消费 672 元。

(2)两次购物有四种情况：

①两次均未打过折，购物总价为 520+600=1120(元)，则实际消费：1000×0. 8+(1120−1000)×0. 6=872(元)；

②600 元打过折，则购物总价为 520+600÷0. 8=1270(元)，则实际消费：1000×0. 8+(1270−1000)×0. 6=962(元)；

③520 元打过折，则购物总价为 520÷0. 8+600=1250(元)，则实际消费：1000×0. 8+(1250−1000)×0. 6=950(元)；

④两次均打折，则购物总价为 520÷0. 8+600÷0. 8=1400(元)，则实际消费：1000×0. 8+(1400−1000)×0. 6=1040(元)。答：此时他的实际消费金额为 872 元或 962 元或 950 元或 1040 元。

(3)600 元打八折后的价格为 600×80%=480(元)，则折后价格一定小于等于 600 元，所以可得：当 $480<x\leq 600$ 时，既可以是打折后的价格，也可以是不打折的价格。

答：当 $480<x\leq 600$ 时，商品的价格既可以打折也可以不打折。

22. 解：(1)4 【解法提示】如解图①所示，第一次剪下边长为 3 的正方形，剩下长为 4、宽为 3 的长方形；第二次剪下边长为 3 的正方形，剩下长为 3、宽为 1 的长方形；第三次剪下边长为 1 的正方形，剩下长为 2、宽为 1 的长方形；第四次剪下边长为 1 的正方形，剩下边长为 1 的正方形，所以它是 4 元理想长方形。

第 22 题解图①

(2)画出长方形 $ABCD$ 及裁剪线的示意图如下：

①如解图②所示：

第 22 题解图②

②如解图③所示：

第 22 题解图③

③如解图④所示；

④如解图⑤所示：

第 22 题解图④ 第 22 题解图⑤

2. 2025 GDFZ 入学数学真卷(二)

快速对答案

一、选择题 1~5 BCADB	
二、填空题 6. 9 7. 32. 4 8. 6 9. 59:286 10. 2. 25	
11. 18 12. 22 13. 5 14. $3\frac{1}{2}$	
计算题、解答题请看"详解详析"P2	

详解详析

一、1. B 2. C 3. A 4. D

5. B 【解析】将总工作量看作"1"，则甲的工效为 $\frac{1}{20}$，乙的工效为 $\frac{1}{12}$，设乙做了 x 天，那么甲先做了(14−x)天，由题可知：$\frac{1}{20}(14-x)+\frac{1}{12}x=1$，解得 $x=9$，那么甲先做了 14−9=5(天)。

二、6. 9 【解析】4※1=3×4−2×1=10，x※10=3x−2×10=7，解得 $x=9$。

7. 32. 4 【解析】瓶子的容积=药水的体积+空余部分的体积，空余部分的体积=25. 2+7×2=7. 2(立方厘米)，所以瓶子的容积=25. 2+7. 2=32. 4(立方厘米)=32. 4 毫升。

8. 6 【解析】因为 10 天长到 32 厘米，9 天长到：32÷2=16(厘米)，8 天长到：16÷2=8(厘米)，7 天长到：8÷2=4(厘米)，6 天长到：4÷2=2(厘米)，所以经过了 6 天竹笋长到 2 厘米。

9. 59:286 【解析】甲瓶中盐含量：2÷(2+9)=$\frac{2}{11}$，甲瓶中水含量：9÷(2+9)=$\frac{9}{11}$，乙瓶盐含量：3÷(3+10)=$\frac{3}{13}$，乙瓶水含量：10÷(3+10)=$\frac{10}{13}$，两瓶混合盐量：$\frac{2}{11}+\frac{3}{13}=\frac{26}{143}+\frac{33}{143}=\frac{59}{143}$，两瓶混合水量：$\frac{9}{11}+\frac{10}{13}=\frac{117}{143}+\frac{110}{143}=\frac{227}{143}$，盐：水=

③Answer

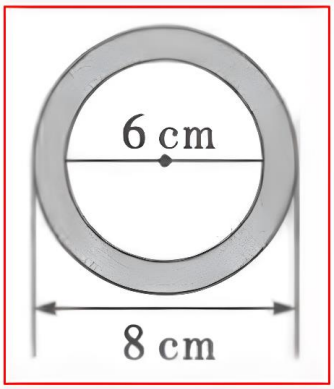
Figure 9: Original answer located in a separate multi-column document.

Unpublished Not for

Question 5

5.[圆环的面积]如图，阴影部分的面积是（ ）平方厘米。（π 取3.14）

① Question



② Figure

A. 6.28 B. 87.92 C. 21.98 D. 314

① Question

Answer: C

③Answer

Solution:

5.C【解析】阴影部分的面积：$3.14 \times \left[(8 \div 2)^2 - (6 \div 2)^2\right] = 21.98$（平方厘米）。解题技巧圆环面积公式：$S = \pi(R^2 - r^2)$

Figure 10: Extracted and reconstructed VQA pair rendered in Markdown.

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Not for dis

B Prompts Used in Dataset Curation

B.1 Identifier-Guided Semantic Reconstruction of VQA Pairs (Section 3.1.2)

The prompt used for LLM to reconstruct VQA Pairs from the PDF layout with block identifiers is as follows:

VQA Pairs Reconstruction

```
You are an expert in {subject}. You are given a json file. Your task is to segment the content, insert images tags, and extract labels:
1. Every json item has an "id" field. Your main task is to output this field.
2. You need to segment the content into multiple `<qa_pair>`...`</qa_pair>` blocks, each containing a question and its corresponding answer with solution.
3. If the problem or answer is not complete, omit them.
4. You need to put the images id into proper positions. You could look at the caption or context to decide where to put the image tags.
5. You will also need to extract the chapter title and each problem's label/number from the text.
6. You only need to output "id" field for chapters, questions and solutions. DO NOT OUTPUT ORIGINAL TEXT. Use ',' to separate different ids.
7. However, use original labels/numbers for labels, and use original numbers for answers. DO NOT output "id" field for labels and answers. You will need to extract
them from the text.

Strict extraction rules:
** About questions and answers/solutions **
- Preserve each problem's original label/number, such as "Example 3", "Exercise 1", "11". Do not include the period after the number. Use Arabic numerals only. For
example, if the label is "IV", convert it to "4".
- If the full label is "III.16", keep only "16". If the full label is "5.4", keep only "4".
- If there are multiple sub-questions (such as "(1)", "(a)") under one main question, always put them together in the same `<qa_pair>`...`</qa_pair>` block.
- If a question and its answer/solution are contiguous, wrap them together as a single `<qa_pair>`...`</qa_pair>` block, e.g.:
  `<qa_pair><label>Example 1</label><question>...</question><answer>...</answer><solution>...</solution></qa_pair>`
- If only questions or only answers with solutions appear, wrap each question or answer with solution in a `<qa_pair>`...`</qa_pair>` block with the missing part
left empty. For example, if only questions appear:
  `<qa_pair><label>Example 1</label><question>...</question><answer></answer><solution></solution></qa_pair>`
- If multiple questions and solutions appear, wrap each question/solution pair in its own `<qa_pair>`...`</qa_pair>` block.
- If you do not see the full solution, only extract the short answer and leave the solution empty. YOU MUST KEEP QUESTIONS WITH ONLY SHORT ANSWERS !!!
** About chapter/section titles **
- Always enclose qa pairs in a `<chapter>`...`</chapter>` block, where <title>MAIN_TITLE</title> is the chapter title or section title.
- Normally, chapter/section titles appear before the questions/answers in an independent json item.
- There could be multiple `<chapter>`...`</chapter>` blocks if multiple chapters/sections exist.
- **Any titles followed by a question/answer whose label/number is not 1, or with a score, should NOT be extracted.**
- Do not use nested titles.
- Leave the title blank if there is no chapter title.
** About figures/diagrams **
- Whenever the question or answer/solution refers to a figure or diagram, record its "id" in question/answer/solution just like other text content.
- You MUST include all images referenced in the question/answer/solution.


If no qualifying content is found, output:
<empty></empty>

Output format (all tags run together, no extra whitespace or newlines except between entries):
<chapter><title>MAIN_TITLE_ID</title>
<qa_pair><label>...</label><question>QUESTION_IDS</question>
<answer>ANSWER(EXTRACTED FROM SOLUTION)</answer><solution>SOLUTION_IDS</solution></qa_pair>
<qa_pair><label>...</label><question>QUESTION_IDS</question>
<answer>ANSWER(EXTRACTED FROM SOLUTION)</answer><solution></solution></qa_pair>
</chapter>
<chapter><title>MAIN_TITLE</title>
<qa_pair><label>...</label><question>QUESTION_IDS</question>
<answer>ANSWER(EXTRACTED FROM SOLUTION)</answer><solution>SOLUTION_IDS</solution></qa_pair>
</chapter>


Example:
<chapter><title>1</title>
<qa_pair><label>Example 1</label><question>2,3</question>
<answer>4/5</answer><solution>5,6,7</solution></qa_pair>
<qa_pair><label>Example 2</label><question>8,9,10</question>
<answer>3.14</answer><solution></solution></qa_pair>
</chapter>
<chapter><title>12</title>
<qa_pair><label>Example 1</label><question>13,14</question>
<answer>2^6</answer><solution>16</solution></qa_pair>
</chapter>

Please now process the provided json and output your result.
```

B.2 Sub-question Decoupling (Section 3.2.1)

The prompt used for breaking down sub-questions in one compound question is as follows:

Sub-question Decoupling

```
You are an educational question structure analysis assistant. Below is a composite question and its corresponding answer. Please split it into several independent sub-questions.
The requirements are as follows:

1. The question may contain multiple sub-questions (e.g., (a)(b)); please accurately identify and split them one by one. Only split sub-questions with clear labels.
   Do not split implicit sub-questions (such as "What is the value of x and y?" or multiple question marks).
2. Each sub-question must be self-contained and answerable. If the original question contains contextual information, include it in each sub-question to preserve full meaning.
   If sub-questions are related (e.g., "(a) Find x. (b) Using the value of x, find y."), do not split them; keep them as one sub-question.
3. If an answer or/and solution is provided, try to match each sub-question with its corresponding part of the answer or/and solution based on semantics.
4. If the original answer or/and solution contains LaTeX formulas, preserve them exactly as they appear.
5. If the original answer or/and solution is missing or cannot be clearly aligned, leave `"sub_answer"` or/and `"sub_solution"` as an empty string.
6. The output must be a valid JSON array, where each element contains:

  * `"sub_id"`: the index of the sub-question (an integer starting from 1)
  * `"sub_question"`: the complete text of the sub-question (or "ORIGINAL" if no splitting is needed)
  * `"sub_answer"`: the corresponding answer, empty string if unavailable (or "ORIGINAL" if no splitting is needed)
  * `"sub_solution"`: the corresponding solution, empty string if unavailable (or "ORIGINAL" if no splitting is needed)

[Important Notice]
1. In some questions, answers or solutions, there will be figures written as `![image](image_url)`. When splitting, please keep these figure references in the corresponding sub-questions, sub-answers, or sub-solutions as EXACTLY what they are.
2. If the question does not need to be split, return an array with a single element, simplified as: [{"sub_id": 1, "sub_question": "ORIGINAL", "sub_answer": "ORIGINAL", "sub_solution": "ORIGINAL"}]
   In this case, you only need to output "ORIGINAL" instead of the full text for sub_question, sub_answer, and sub_solution, so that we can save tokens.

## Example Input:

**Question:**
A class has 40 students, including 25 boys and 15 girls. ![image](question_images/a284h5iuh38.jpg) (a) Find the percentage of boys in the class. (b) Find the percentage of girls in the class.

**Answer:**
(a) 62.5%. (b) 37.5%.

**Solution:**
Percentage of boys = (25/40) * 100 = 62.5%, percentage of girls = (15/40) * 100 = 37.5%.
----------------------------------------------

## Example Output:

```json
[
  {
    "sub_id": 1,
    "sub_question": "A class has 40 students, including 25 boys and 15 girls. ![image](question_images/a284h5iuh38.jpg) Find the percentage of boys in the class.",
    "sub_answer": "62.5%.",
    "sub_solution": "Percentage of boys = (25/40) * 100 = 62.5%."
  },
  {
    "sub_id": 2,
    "sub_question": "A class has 40 students, including 25 boys and 15 girls. ![image](question_images/a284h5iuh38.jpg) Find the percentage of girls in the class.",
    "sub_answer": "37.5%.",
    "sub_solution": "Percentage of girls = (15/40) * 100 = 37.5%."
  }
]
```
Now, please split the following question according to the above requirements:
[Question]
{input_question}

[Answer]
{input_answer}

[Solution]
{input_solution}
```

B.3 Problem Type Classification (Section 3.2.2)

The prompt used for classifying problem types is as follows:

Type Classification

```
[Role]
You are an education expert familiar with textbook question formats at high school and university levels.
Your task is to determine the question type based on the question and answer provided.

[Possible Categories]
Choose exactly one of the following types:
1. Proof problem - requires proving a statement, identity, inequality, or property.
2. Explanation problem - asks for reasoning, causes, interpretation, principle, or conceptual explanation.
3. Fill-in problem - asks to fill in blanks, complete missing expressions, or supply intermediate steps.
4. Calculation problem - involves explicit numerical or symbolic computation, formula manipulation, or value derivation.
Even if the final answer is a short conclusion such as "thus xxx increases" or "so the velocity decreases,"
it should still be considered a Calculation problem if the majority of the reasoning is computational.
5. Multiple-choice problem - asks to choose or identify the correct option (e.g., "Which of the following").
6. Sketching/Plotting problem - requires sketching a figure, diagram, graph, or geometric representation.
7. Other - for tasks that don't fit any of the above types.

[Judgment Rules]
1. If the problem explicitly says "prove," "show that," "derive," -> classify as Proof problem.
2. If it mainly contains explanations, reasoning, or conceptual analysis without detailed calculation -> Explanation problem.
3. If the question has blanks, missing terms, or placeholders (e.g., "( )" or "____") -> Fill-in problem.
4. If there are multiple formula derivations, substitutions, or numeric results -> Calculation problem,
even if followed by a brief explanatory conclusion.
5. If it asks to select the correct answer among options (A/B/C/D, etc.) -> Multiple-choice problem.
6. If the question explicitly requires producing a figure, diagram, plot, or geometric construction -> Sketching/Plotting problem.
7. If none of these clearly apply or the problem type is mixed -> Other.

[Output Format]
Return a JSON object with the following fields:
{
  "type": "Calculation | Proof | Explanation | Fill-in | Multiple-choice | Sketching/Plotting | Other",
  "reason": "Brief justification for the classification."
}

Please determine the type of the following question and output only one of the above category names.
(Proof, Explanation, Fill-in, Calculation, Multiple-choice, Sketching, Other).

[Question]
{input_question}

[Answer]
{input_answer}
```

B.4 Semantic Answer Filtering (Section 3.2.4)

The prompt used for semantic answer extraction is as follows:

Semantic Answer Extraction

```
[Role]
You are a professional question answering system. Extract concise and accurate answers from the provided solutions.
Output only the answer without any additional text.

Extract the answer from the following solution:

Solution: {solution}
```

B.5 Quality Filtering (Section 3.2.4)

The prompt used for data quality filtering is as follows:

Quality Filtering

```
[Role]
You are an education expert familiar with textbook question formats at high school and university levels.
Your task is to determine whether the provided question and answer pair is suitable to serve as a problem in an exam.

Question: {input_question}

Answer: {input_answer}

[Criteria]
```

```
1. Clarity: The question must be suitable for an exam setting, meaning it should raise **a clear problem** that requires a specific solution.
Examples, **statements without questions**, open-ended discussions and other context that do not pose a clear problem are not suitable.
Questions like "Give an example of..." that can have many valid answers are also not suitable.
You should be particularly careful with questions that **only provide a topic or theme** without a specific problem to solve.
For instance, "all primes less than 100" is not a valid question, because it does not specify what to do (listing, counting, ...) with those primes.
Instead, a question like "List all primes less than 100" or "How many primes are there less than 100?" would be suitable.
2. Relevance: The answer must directly address the question asked.
If the answer seems to be addressing a different question and is wrongly paired with the given question, it is not suitable.
3. Completeness and Self-Containment: The question and answer should be complete and self-contained, providing all necessary information for understanding and
solving it without requiring external context.
Questions that rely heavily on prior context or external references are not suitable.
Answers such as "Refer to theorem X", "Corollary of previous result", "Answered in the text above", "Omitted for brevity" are not acceptable.
Incomplete questions or answers that leave out critical information are also not suitable.
4. Explicit Task Requirement: The question must contain an explicit task phrase (such as "compute", "determine", "find", "prove", "list", "show", "give the value of
", etc.).
Pure expressions or noun phrases are NOT acceptable even if they are commonly understood as implicit tasks in mathematical contexts.
If the question does not include an explicit verb specifying what the student must do, it must be judged unsuitable.
Of course, if the question is in a multiple-choice or fill-in-the-blank format, the choices or blanks themselves will serve as the explicit task requirement.

[Important Notice]
1. You do not need to evaluate the correctness of the answer, only whether it is appropriate and complete in relation to the question.
2. Short answer with no explanation (calculation, proof, counterexample, ...) is acceptable as long as it directly addresses the question.
3. You should be very strict in your evaluation. If any of the criteria above are not fully met, the question-answer pair should be considered unsuitable.

[Output Format]
Return a JSON object with the following fields:
{
"reason": "Brief justification of your judgement."
"judgement": "true | false",
}

Your judgment:
```

C Training and Evaluation Details

C.1 Detailed Training Hyperparameters

Table 4 lists the hyperparameters used in SFT of Qwen3-8B-Base in our experiments.

Table 4: Detailed Hyperparameters for SFT of Qwen3-8B-Base Instruct.

HyperParameter	Value
Method	Full Fine-tuning
Learning Rate	5×10^{-5}
Learning Rate Scheduler	Cosine
Warmup Ratio	0.1
Training Epochs	3.0
GPUs	8
Per-GPU Batch	2
Gradient Accumulation Steps	4
Effective Batch Size	64
Optimizer	AdamW
Validation Split Size	0.1
Max Sequence Length	4096
Numerical Precision	BF16

Table 5 lists the hyperparameters used in SFT of Qwen3-VL-8B Instruct in our experiments.

C.2 Answer Generation and Verification Prompts

The answer generation prompt used for both distillation, training and evalution is as follows:

Answer Generation

```
You are an intelligent chatbot designed for producing the answer to the given reasoning task.
Remember: DO NOT output anything else, only output the answer you generate.
Generate a solution to the given task strictly following this format:
```

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Table 5: Detailed Hyperparameters for SFT of Qwen3-VL-8B Instruct.

HyperParameter	Value
Learning Rate	5×10^{-6}
Learning Rate Scheduler	Cosine
Warmup Steps	5
Training Epochs	2.0
Batch Size	128
Optimizer	AdamW
Maximum Gradient Norm	1.0
Vision Tower	Frozen
Multi-modal Projector	Trainable
Validation Split Size	0.05
Max Sequence Length	12,800
Image Max Pixels	589, 824
Video Max Pixels	65, 536
Numerical Precision	FP16

draft.

```
1. Identify key components and premises of the task
2. Apply relevant principles, theorems, or methods with step-by-step derivation or argument
3. Perform any necessary calculations or logical checks with intermediate verification
4. Present the final answer or conclusion in a clear, unambiguous notation

Format Requirements:
- Prefix each step with arrow symbol
- Ensure all symbols and special characters are presented using appropriate markup (e.g., LaTeX commands for mathematical symbols, code formatting for code snippets)

Example Template:
Task: Analyze the time complexity of the following sorting algorithm and prove its correctness.

Solution:
1. Identify components:
-> Algorithm uses divide-and-conquer to split the list in half
-> Merging step compares elements pairwise

2. Apply principles:
-> Recurrence: T(n) = 2T(n/2) + O(n)
-> By Master Theorem, T(n) = O(n log n)

3. Verification:
-> Check base case T(1) = O(1)
-> Inductive step holds for n = 2^k

4. Conclusion:
-> The algorithm runs in \\boxed{O(n\\log n)} time and correctly sorts any input list.

Here is the given task you need to solve:
{question}
```

The prompt for answer verification is as follows:

Answer Verification

```
As an answer evaluation expert, please assess whether the following answer is correct.

Question: {question}

Reference Answer: {reference_answer}

Current Answer: {answer}

Please carefully analyze whether the current answer is semantically consistent with the reference answer.
Focus only on comparing the answers themselves, not on how the problem is solved.
Don't just look at the surface text, understand the essential content of the answers.
If the current answer is semantically consistent with the reference answer, even if expressed differently, it should be judged as correct.
For numerical calculation problems, also consider whether the answer is within the acceptable error range (typically 5%). Be careful to differentiate whether the
question is indeed a numerical calculation or one that requires a strictly identical answer.
```

```
Please return your judgment result in JSON format:
{{"judgement_result": true}} indicates the answer is correct
{{"judgement_result": false}} indicates the answer is incorrect

Your judgment:
```

D Evaluation Guidelines for Human Annotators on VQA Extraction

The goal of this evaluation is to verify the structural and content integrity of VQA pairs extracted from PDFs, mainly focusing on the performance of the LLM's reconstruction logic.

D.0.1 Precision Errors (Incorrect Extractions). Mark the sample as **incorrect** if any of the following extra or wrong information is present:

- **QA Mismatch**: The extracted question does not correspond to the provided answer.
- **Hallucinations**: The text in questions or answers contains contents that do not exist in the original source document.
- **Ordering Issues**: QA pairs are not extracted in the proper sequence as they appear in the document.
- **Image Misplacement**: Images are inserted at the wrong location (e.g., an image related to Question 5 is placed inside Question 2, or an image related to Answer 2 is is placed inside Question 2).

D.0.2 Recall Errors (Missing Information). Mark the sample as **incorrect** if information from the source is omitted:

- **Incompleteness**: A QA pair is missing a significant portion of its required text (e.g., half of the answer is cut off), or the whole question/answer is missing.
- **Missing Images**: An image related to the question or answer in the PDF is not inserted into the extracted output.

D.0.3 Acceptable Deviations. You may mark the sample as **correct** even if the following minor issues are present:

- **Text-level Parsing Glitches**: Minor character errors or formatting artifacts that do not change the fundamental meaning of the sentence.
- **Formula Errors**: Complex mathematical LaTeX or symbols that are incorrectly rendered but do not affect understanding.

E Benchmark Details

E.0.1 QA Benchmarks. We evaluate on nine QA benchmarks covering general reasoning, mathematical reasoning, and physics reasoning.

- **MMLU** [9]: A multidisciplinary benchmark covering 57 subjects across STEM, humanities, and social sciences. We use the official test set (14k questions).
- **MMLU-Pro** [27]: An enhanced version of MMLU designed to improve evaluation difficulty and discriminative power. We use the official test set (12k questions).
- **BBH** [24]: BIG-Bench Hard consists of 23 challenging reasoning tasks evaluating logical reasoning, symbolic manipulation, and complex arithmetic. We use the full dataset (6.5k questions).
- **SuperGPQA** [5]: A graduate-level knowledge reasoning benchmark with broad disciplinary coverage. We use the official test set (26.5k questions).
- **GSM8K** [3]: A grade-school mathematical reasoning benchmark consisting of multi-step arithmetic word problems. We use the official test set (1k questions).
- **AIME**: Competition-level mathematics problems from the 2024 and 2025 American Invitational Mathematics Examination. The evaluation set contains 60 problems.
- **MATH500** [10]: A challenging subset of the MATH dataset covering algebra, geometry, number theory, calculus, probability, and statistics. We use the full evaluation set (500 problems).
- **UGPhysics** [28]: An undergraduate-level physics QA benchmark covering mechanics, electromagnetism, thermodynamics, optics, and modern physics. We use the official test set (5.5k questions).

E.0.2 VQA Benchmarks. We further evaluate multimodal reasoning on seven VQA benchmarks, including our in-domain test split and six out-of-distribution datasets.

- **FlipVQA-test**: A held-out test split of FlipVQA containing 998 problems sampled from the full dataset.
- **MMMU** [31]: A multidisciplinary multimodal benchmark designed to evaluate college-level knowledge and reasoning. We use the validation set (900 problems).
- **OlympicArena** [12]: An Olympiad-level multimodal benchmark covering seven disciplines. We use the validation split and retain only VQA problems, resulting in 268 evaluation instances.
- **MathVerse** [33]: A visual mathematics benchmark containing problems with varying degrees of visual information. We use the *testmini* split excluding text-only instances (3.94k problems).
- **MathVista** [18]: A benchmark for mathematical reasoning in visual contexts, including plots, diagrams, and scientific figures. We use the *testmini* split (1k problems).

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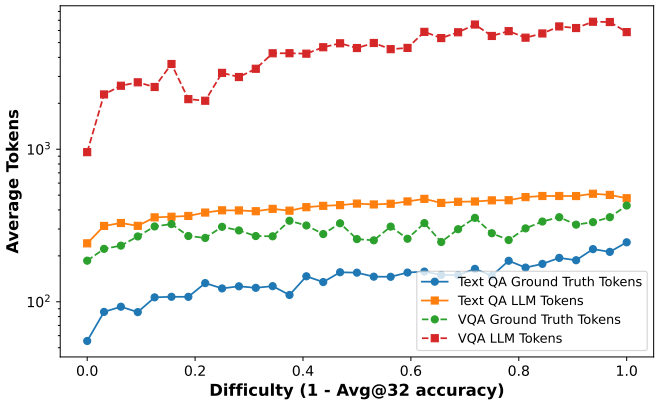

Figure 11: Average tokens generated for QA and VQA tasks. Visual Question Answering (VQA) requires significantly more tokens from the LLM (red line) compared to both its ground truth and text-only tasks, especially as task difficulty increases.

- **OlympiadBench** [8]: An Olympiad-level bilingual multimodal benchmark covering mathematics and physics. We exclude text-only and proof-based problems, resulting in 3.68k evaluation instances.
- **MatSciBench** [32]: A multimodal benchmark for materials science question answering. We use the test split excluding text-only problems (315 instances).

F Observations

F.1 Difficulty-induced Token Expansion

We observe a distinct positive correlation between task difficulty and output length (as shown in Fig. 11). Notably, this "token expansion" is disproportionately magnified in VQA, where responses often exceed 6,000 tokens—an order of magnitude longer than in QA. This suggests that multimodal reasoning requires extensive "perceptual-to-logical" translation, where exhaustive visual grounding must precede logical inference. Consequently, long-form CoT emerges as a structural hallmark of complex multimodal tasks.

F.2 Impact of Training on High-Difficulty Samples

Given that difficulty scales with reasoning length, we evaluate fine-tuning on high-difficulty subsets ($Difficulty \geq 0.4$). As shown in Table 6, prioritizing these samples enhances performance on intensive reasoning benchmarks such as OlympicArena and OlympiadBench. This suggests that high-difficulty data provides the gradient density essential for deep reasoning optimization, despite a slight trade-off in general knowledge breadth on broader benchmarks like MMMU.

Table 6: Performance evaluation of models fine-tuned on the *Full* FlipVQA dataset and its difficult subset across various reasoning benchmarks. *Difficulty≥ 0.4* subset keeps the problems on which the accuracy of Qwen3-VL-8B-Instruct is less than 0.6 after rolling out 32 times.

Benchmark	Flip-VQA Test	MMMU	OlympicArena	MathVerse	MathVista	OlympiadBench	MatSciBench
Full Dataset	32.97	**66.22**	47.39	**71.37**	70.10	64.14	**35.62**
Difficulty≥ 0.4	**33.77**	65.33	**49.25**	71.19	**71.00**	**66.42**	33.56

F.3 Effect of CoT Length in Training Data

To investigate the impact of CoT length, we selected samples from the mathematics CoT dataset whose lengths are above the median and used them for fine-tuning. As shown in Table 7, the results indicate that training with longer CoT data improves the model's reasoning ability, particularly on more challenging benchmarks such as AIME2024 and AIME2025.

F.4 Impact of Modality Mixing

We also study the impact of mixing QA and VQA data during training. To avoid imbalance in the data distribution, we sample a subset of the QA dataset that matches the size of the VQA dataset. We evaluate five data composition strategies: (1) VQA-only; (2) QA-only; (3) Sequential (VQA → QA); (4) Sequential (QA → VQA); and (5) Joint Training (VQA + QA). The results in Table 8 reveal several key observations:

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Table 7: Performance of Qwen3-8B-Base fine-tuned with data of different CoT lengths on several math benchmarks. † Long CoT data are the samples from the math CoT dataset whose lengths are above the median.

Benchmark	GSM8K	MATH500	AIME2024	AMIE2025	AMIE2026
Math Data (All)	**93.71**	82.80	18.44	16.98	14.90
Math Data (Long CoT †)	93.40	**84.80**	**22.71**	**20.52**	**21.56**

(1) **Cross-modal transfer**. The **QA-only** setting yields performance gains over the baseline on several VQA benchmarks, suggesting that improvements in textual reasoning can partially transfer to visual reasoning tasks.
(2) **Catastrophic forgetting under sequential training**. In sequential training regimes, the **QA → VQA** strategy underperforms the VQA-only setting, while the **VQA → QA** sequence nearly regresses to the QA-only baseline. This behavior indicates substantial forgetting of visual understanding when the model is subsequently fine-tuned on purely textual data.
(3) **Joint training synergy**: The joint modality setting (VQA + QA) consistently achieves the best performance, demonstrating that textual reasoning improvements can complement visual understanding when both modalities are trained jointly.

Table 8: Performance across different modality data mixture methods. We explore the effects of five distinct data composition strategies. Bold and underlined values indicate the best and second-best results, respectively.

Benchmark	Flip-VQA Test	MMMU	OlympicArena	MathVerse	MathVista	OlympiadBench	MatSciBench
Baseline	24.45	65.78	42.54	66.73	<u>70.30</u>	57.35	29.79
VQA-only	<u>32.97</u>	<u>66.22</u>	**47.39**	<u>71.37</u>	70.10	64.14	<u>35.62</u>
QA-only	28.36	63.11	45.90	68.88	68.40	62.92	34.25
VQA → QA	30.66	64.89	45.52	68.76	68.90	62.70	34.25
QA → VQA	32.46	65.44	46.27	70.71	**72.40**	**65.66**	33.56
VQA + QA	**33.57**	**66.44**	<u>47.01</u>	**72.34**	70.10	<u>64.42</u>	**35.96**

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F.5 Subject-Specific and Cross-Discipline Generalization

We evaluate the impact of domain-specific versus joint fine-tuning across math and physics subsets (Table 9). Our analysis yields two core insights:

- **Cross-domain generalization.** Models trained on physics data demonstrate competitive performance on math benchmarks, suggesting that reasoning patterns learned in physics effectively transfer to math tasks.
- **Cross-discipline synergy:** Joint fine-tuning on both math and physics data consistently outperforms domain-specific training, indicating that exposure to diverse CoT patterns fosters better generalization and mitigates overfitting.

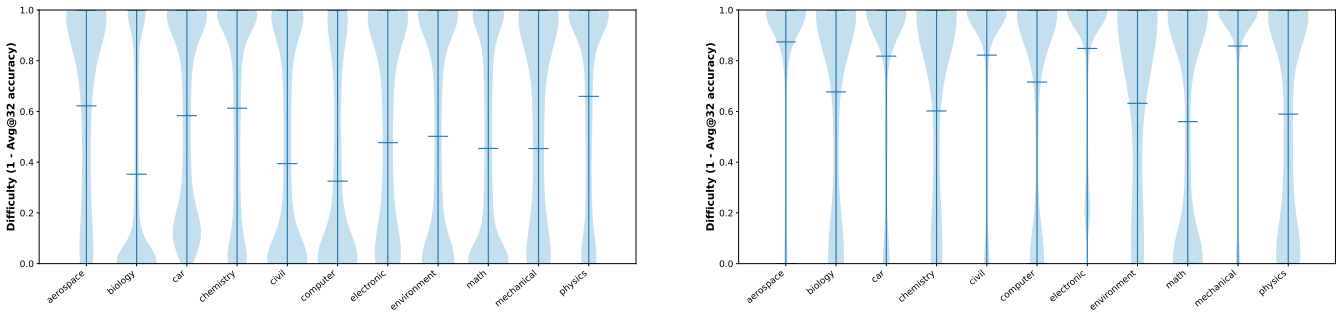
Table 9: Performance of Qwen3-8B-Base fine-tuned with data from different disciplines on several mathematical benchmarks.

Benchmark	GSM8K	MATH500	AIME2024	AIME2025	AIME2026
Math	93.71	82.80	18.44	16.98	14.90
Physics	93.55	82.20	20.73	16.88	13.02
Math + Physics	**93.86**	**84.20**	**21.56**	**22.71**	**17.08**

F.6 Difficulty Distribution

We evaluate question difficulty based on model success rates across multiple rollouts. As shown in Fig. 12, the difficulty exhibits a pronounced bimodal distribution: questions tend to cluster at the extremes (0 or 1), indicating that the model either masters the underlying concept or fails consistently. This "all-or-nothing" pattern is consistent across all subjects. Furthermore, Fig. 12 reveals that domain-specific engineering (e.g., Aerospace) and hard sciences (e.g., Physics) present higher challenges compared to Mathematics. Notably, VQA pairs consistently yield higher difficulty scores than their QA counterparts, highlighting the added complexity of cross-modal reasoning.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009



(a) Text-only QA

(b) VQA pairs

Figure 12: Difficulty distributions across different subjects for text-only QA (left) and VQA pairs (right) in our FlipVQA dataset. Each violin plot illustrates the kernel density estimation of difficulty scores, revealing a consistent bimodal pattern across disciplines.

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